

# LAB EXPERIMENT RECORD

## Set 5: Website Analytics

|                           |                                          |
|---------------------------|------------------------------------------|
| Department                | AIDS                                     |
| Course Code / Course Name | DSA0602/ Data Handling and Visualization |
| Experiment                | Data Visualization using R               |
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| Register Number           | 192424049                                |
| Date                      | 01/10/2026                               |
| Day                       | Thursday                                 |

### Aim

To analyze product inventory data using R programming by creating a bar chart, stacked bar chart, and scatter plot, and to develop a Tableau dashboard for interactive exploration of product quantities and category-wise inventory distribution.

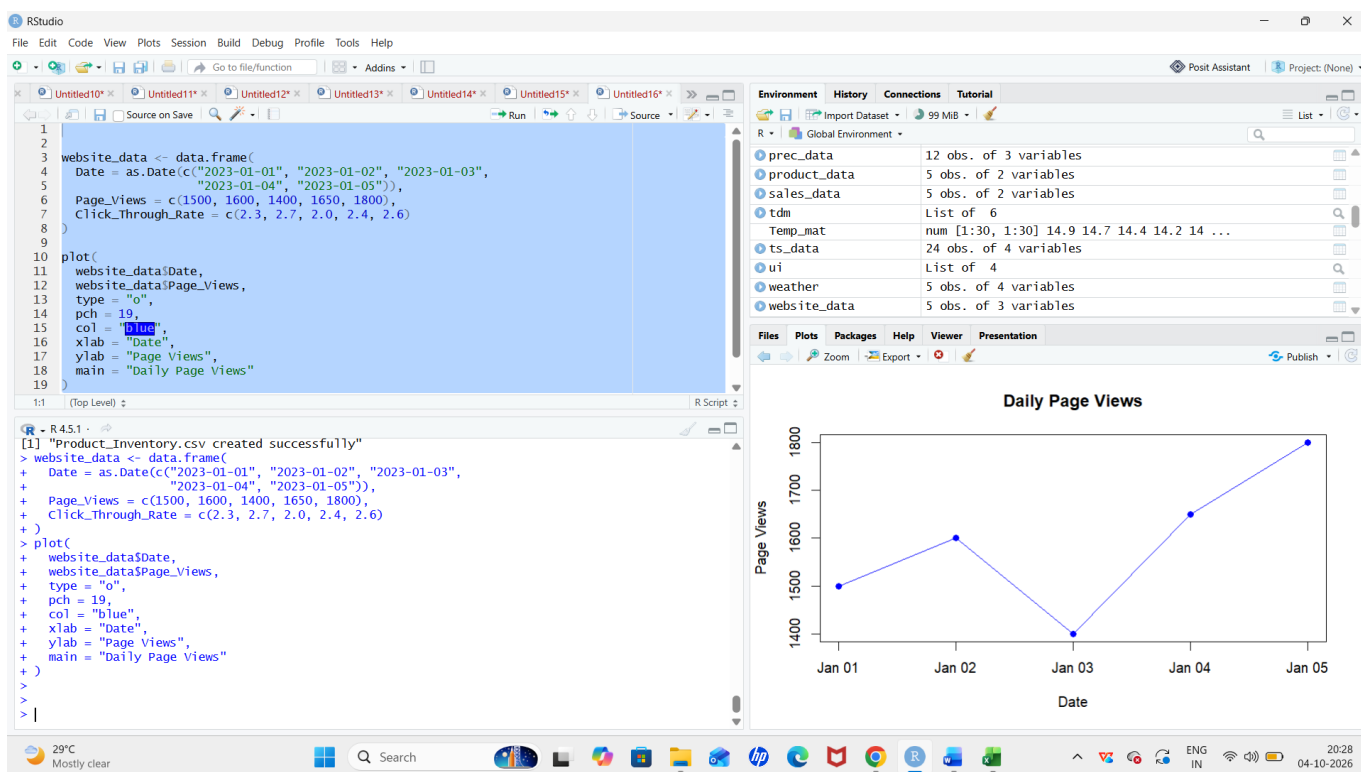
| Date       | Page Views | Click-through Rate |
|------------|------------|--------------------|
| 2023-01-01 | 1500       | 2.3%               |
| 2023-01-02 | 1600       | 2.7%               |
| 2023-01-03 | 1400       | 2.0%               |
| 2023-01-04 | 1650       | 2.4%               |



| Date       | Page Views | Click-through Rate |
|------------|------------|--------------------|
| 2023-01-05 | 1800       | 2.6%               |

1. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the

```
website_data <- data.frame(  
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",  
                  "2023-01-04", "2023-01-05")),  
  Page_Views = c(1500, 1600, 1400, 1650, 1800),  
  Click_Through_Rate = c(2.3, 2.7, 2.0, 2.4, 2.6)  
)  
  
plot(  
  website_data$Date,  
  website_data$Page_Views,  
  type = "o",  
  pch = 19,  
  col = "blue",  
  xlab = "Date",  
  ylab = "Page Views",  
  main = "Daily Page Views"  
)
```



2. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart

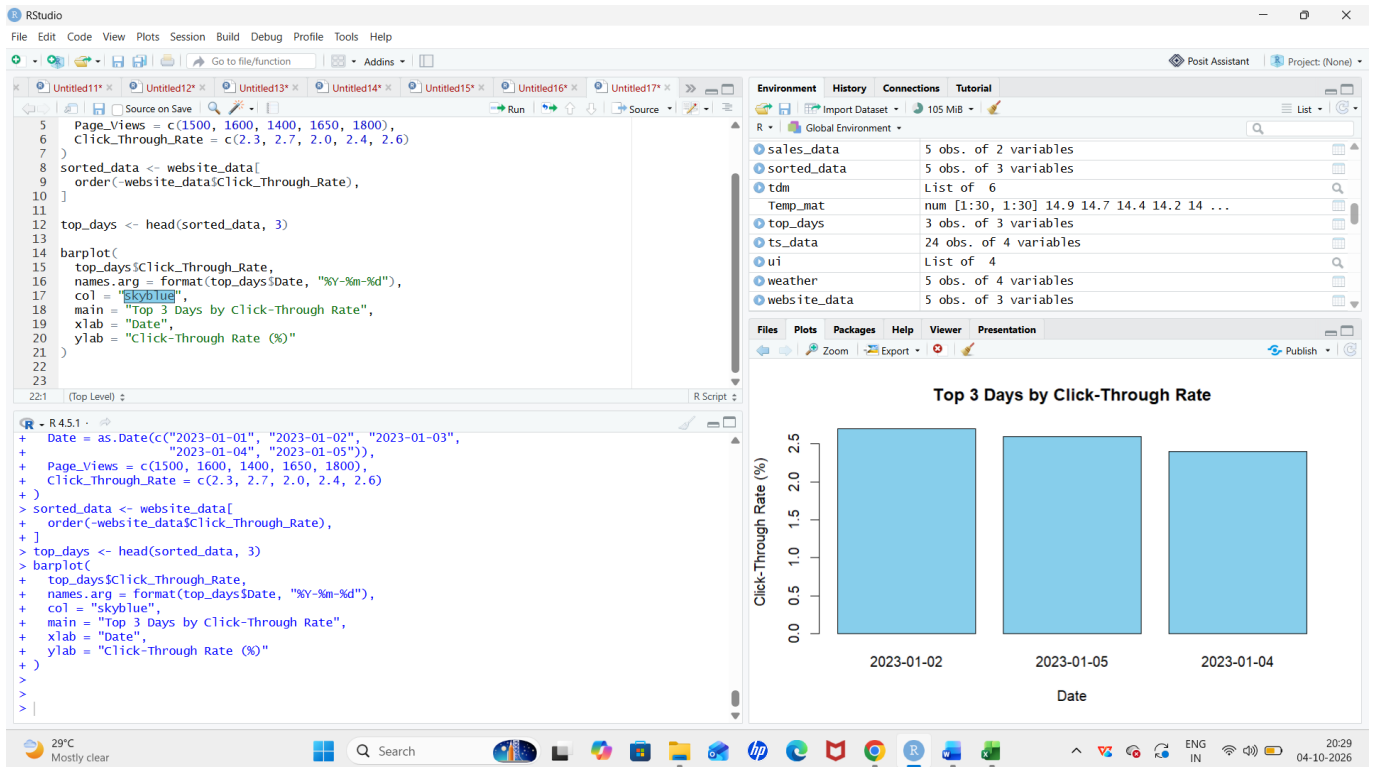
```
# Website Traffic Dataset

website_data <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",
                  "2023-01-04", "2023-01-05")),
  Page_Views = c(1500, 1600, 1400, 1650, 1800),
  Click_Through_Rate = c(2.3, 2.7, 2.0, 2.4, 2.6)
)

# Sort data by Click-Through Rate
sorted_data <- website_data[
  order(-website_data$Click_Through_Rate),
]

# Select top 3 days
top_days <- head(sorted_data, 3)

# Bar Chart
barplot(
  top_days$Click_Through_Rate,
  names.arg = format(top_days$Date, "%Y-%m-%d"),
  col = "skyblue",
  main = "Top 3 Days by Click-Through Rate",
  xlab = "Date",
  ylab = "Click-Through Rate (%)"
)
```



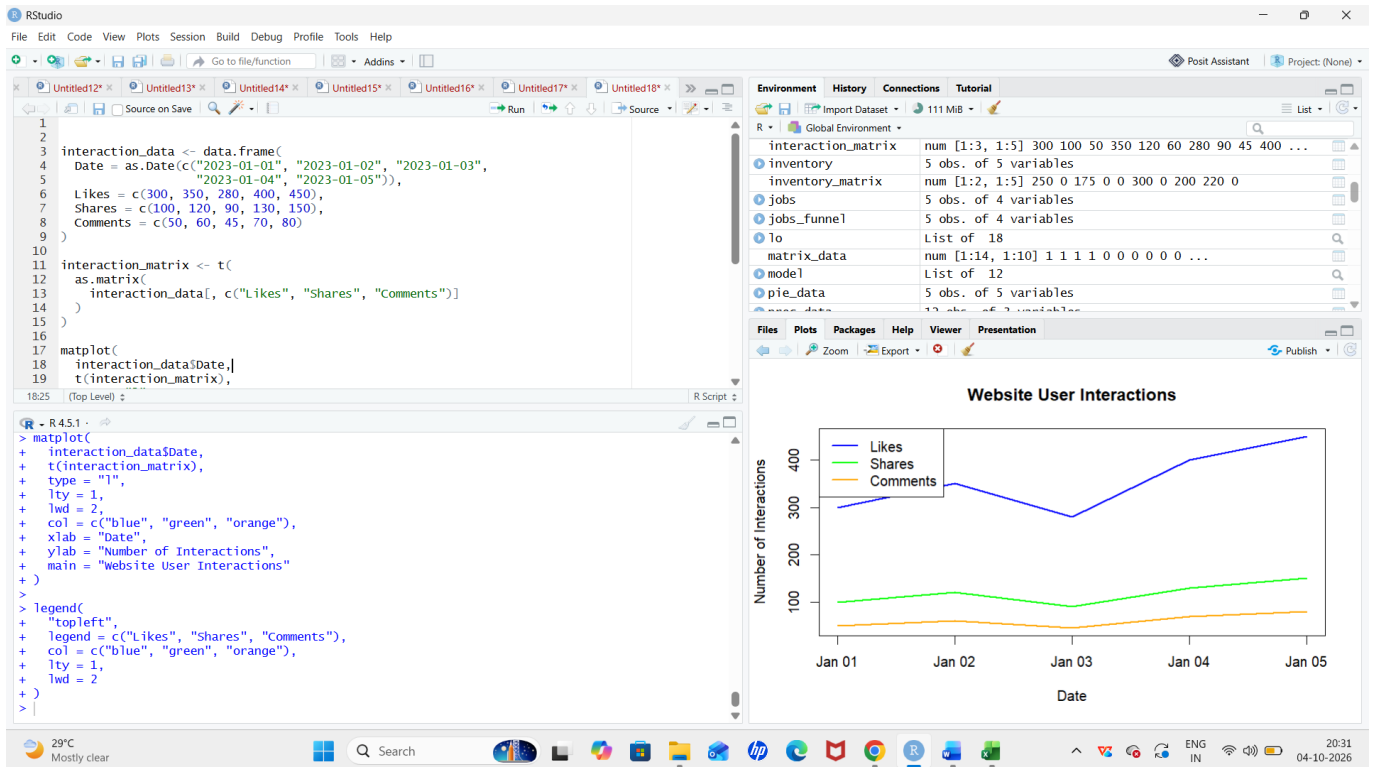
3. Develop a stacked area chart to display the distribution of user interactions (likes, shares, comments) on a website..

```
interaction_data <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",
                  "2023-01-04", "2023-01-05")),
  Likes = c(300, 350, 280, 400, 450),
  Shares = c(100, 120, 90, 130, 150),
  Comments = c(50, 60, 45, 70, 80)
)

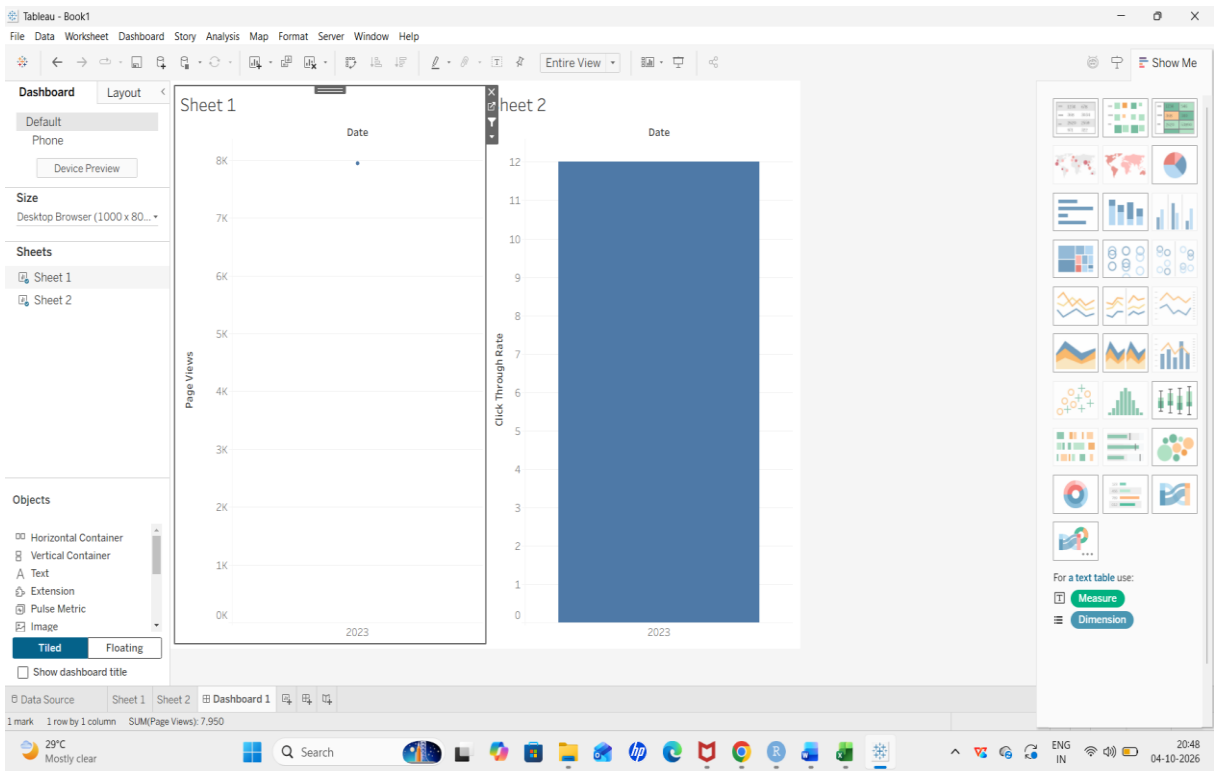
# Create stacked area chart
interaction_matrix <- t(
  as.matrix(
    interaction_data[, c("Likes", "Shares", "Comments")]
  )
)

matplot(
  interaction_data$Date,
  t(interaction_matrix),
  type = "l",
  lty = 1,
  lwd = 2,
  col = c("blue", "green", "orange"),
  xlab = "Date",
  ylab = "Number of Interactions",
  main = "Website User Interactions"
)

legend(
  "topleft",
  legend = c("Likes", "Shares", "Comments"),
  col = c("blue", "green", "orange"),
  lty = 1,
  lwd = 2
)
```



4. Build a Tableau dashboard with the line chart and bar chart for interactive exploration of website traffic data.



## PROCEDURE

1. Create the website traffic dataset in R with Date, Page Views, and Click-through Rate.
2. Convert the Date column into the appropriate date format using the `as.Date()` function.
3. Use the `plot()` function to create a line chart showing the trend in daily page views over time.
4. Add appropriate axis labels and title to the line chart.
5. Arrange the click-through rate values in descending order to identify the top N days with the highest click-through rates.
6. Use the `barplot()` function to create a bar chart showing the top days with the highest click-through rates.
7. Create a dataset containing likes, shares, and comments for website user interactions.
8. Use the appropriate R plotting function to create a stacked area chart showing the distribution of likes, shares, and comments over time.
9. Import the website traffic dataset into Tableau.
10. Create a line chart in Tableau to display daily page views.
11. Create a bar chart in Tableau to display the top days based on click-through rate.
12. Combine both charts into a Tableau dashboard and add filters for interactive exploration.

## RESULT

The website traffic data was successfully analyzed using R programming. The line chart displayed the trend in daily page views, while the bar chart showed the days with the highest click-through rates. The stacked area chart represented the distribution of user interactions such as likes, shares, and comments. Finally, a Tableau dashboard was developed by combining the line chart and bar chart, allowing users to interactively explore website traffic data.



